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GE 1502
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SW3 – Chessboard

Introduction:

For this assignment, a chessboard is being made with SolidWorks. The theme I chose is the video game series “Kingdom Hearts” from the video game publisher Square Enix. In the game, many of the characters and organizations have symbols that represent/classify them. Each piece will be represented by a symbol which represents many of the enemies/characters in the series.

Bulleted List:

- Pawn: Heartless (Smallest NPCs)



- King: Sora (main character)



- Queen: Kairi (main character romantic interest)



- Bishops: Nobodies (larger NPCs, strategic)



- Knights: Unversed (Big NPCs, coming unpredictably)



- Rooks: Moon Shaped Heart (most recognizable symbol in series)



Overview of Used Techniques:

- 3-point arc (King, used for the non-complete circles at the top)
- Spline (many pieces, overall shape of them)
- Polygon (Queen, used to make the circular out cuts)
- Mirror Entities (many pieces, used for symmetry)
- Lofts (Rook, connection from the circle to the rectangle)

- Revolved Boss/Base (Bishop, base for the 2 semi circles)
- Fillet (Pawn, bottom and center of the pawns)
- Draft (Knight, used for the circular base)
- Shell (Rook, shown in the triangular extensions)

Techniques Used In Pawn:

- Spline (used for mainly the outer)
- Mirror (used for the center stripes and thorns)
- Fillet (used in bottom and center)
- Extrude (for outside and inside)
- Offset Entities (to make outside)
- Scale
- Line
- Centerline

The symbol was then assembled onto the base in the same assembly as the chessboard.

Techniques Used in King:

- Mirror
- Extrude
- 3 point arc
- Sketch Picture
- Scale

Techniques Used in Queen:

- Mirror
- Sketch
- Polygon
- Sketch picture
- Extrude
- Cut
- Scale
- Spline
- Circular Sketch Pattern

The symbols were then assembled onto the base in the same assembly as the chessboard

Techniques Used in Rook:

- Extrude
- Spline
- Sketch Image
- Mirror
- Loft (circle to square)

- New Plane
- Shell (base)
- Scale

Techniques Used in Bishop:

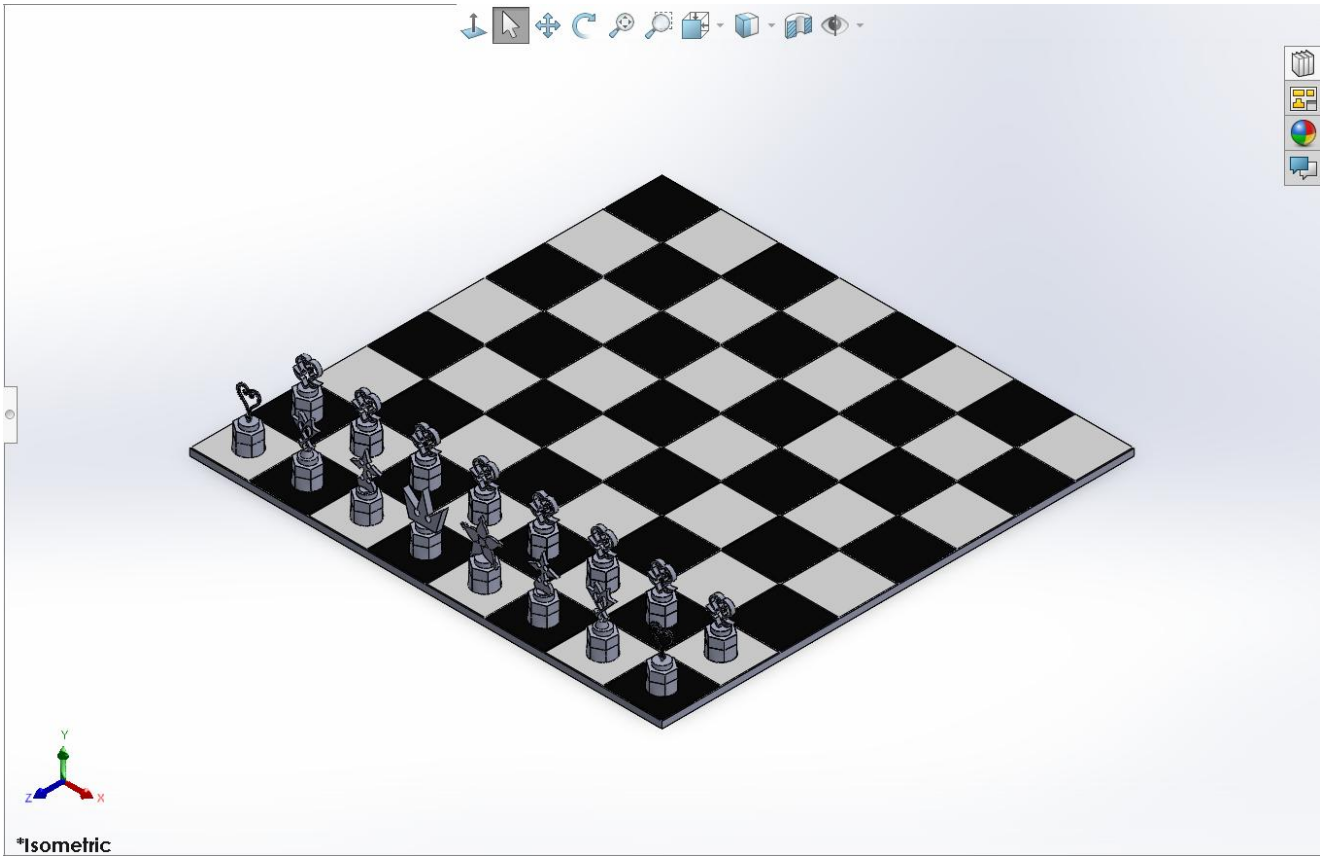
- Extrude
- Sketch Image
- Spline
- Revolved Boss/Base (For the base of the piece)
- Scale

Techniques Used in Knight:

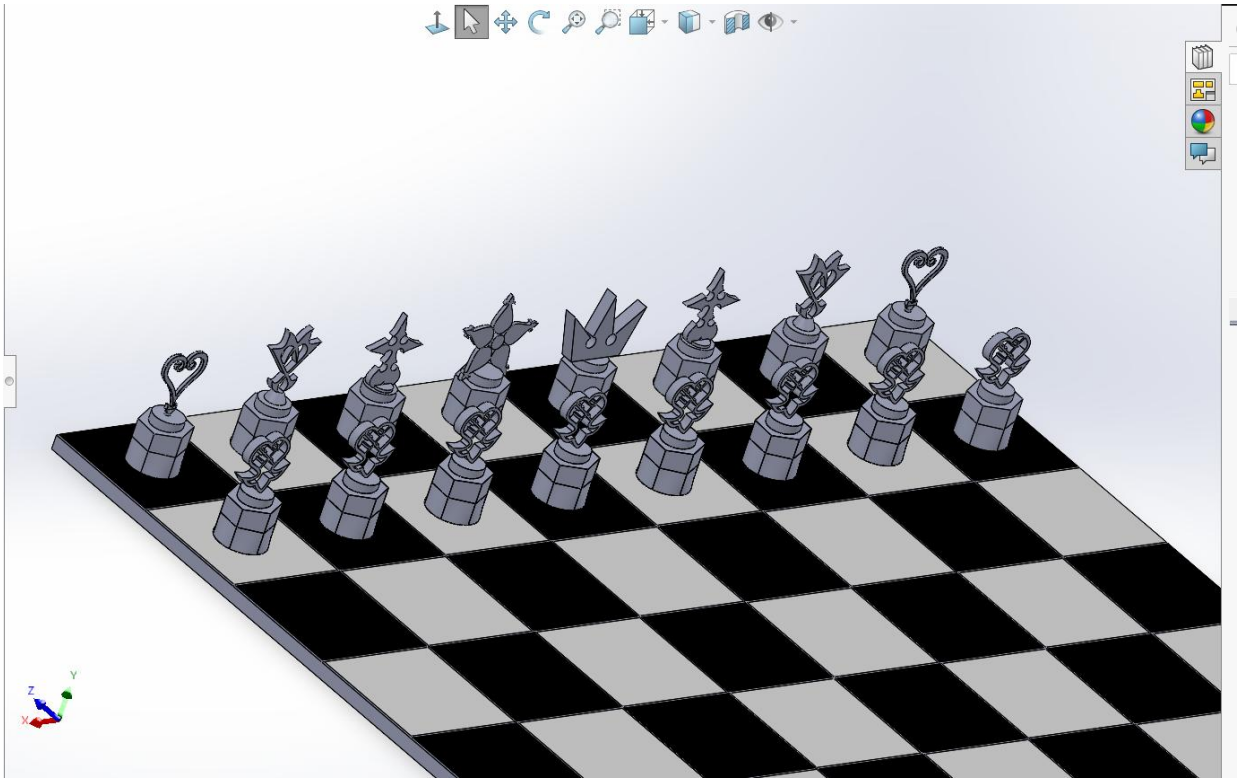
- Draft (to make the circular base that converges in)
- Extrude
- Spline
- Mirror
- Scale

Screenshots:

Isometric of Board

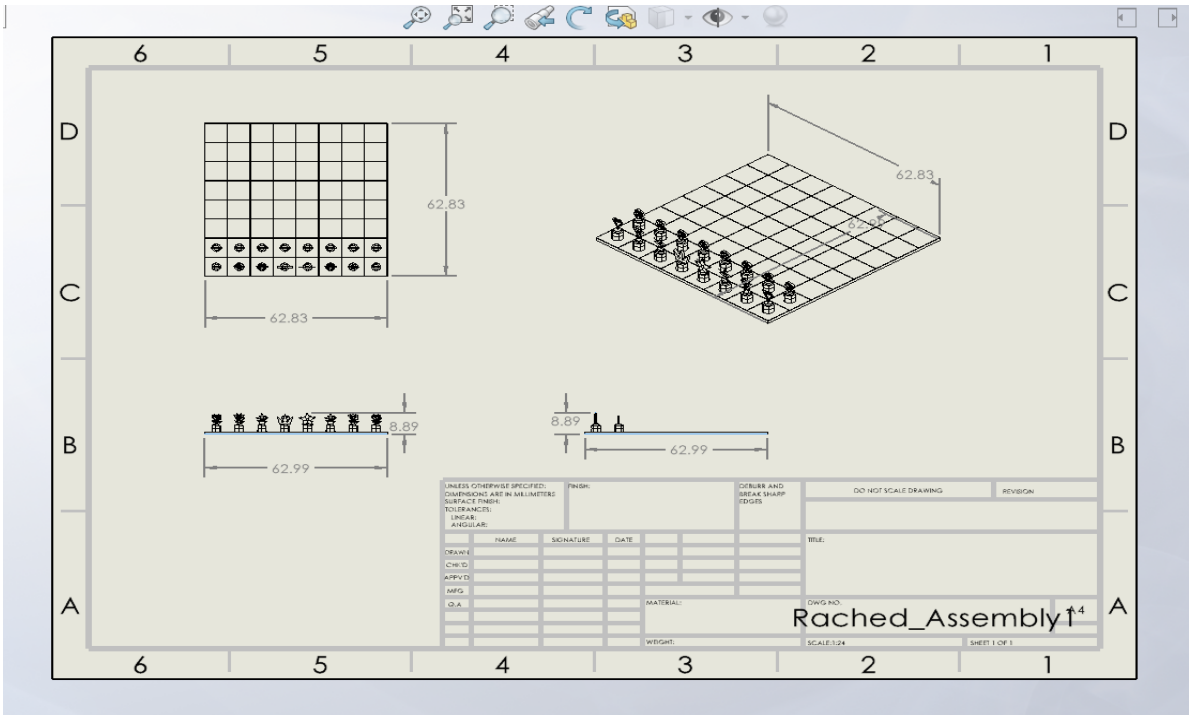


Screenshot of my choice:



Appendix:

Drawing file pictures:



Drawing file of favorite piece:

